

Market Simulation Report

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Simulation Setup

Parameter	Value
Total Agents	100
Simulation Time	30 minutes
Snapshot Interval	1 second
Random Seed	42
Arrival Rate	0.1 events/second/agent

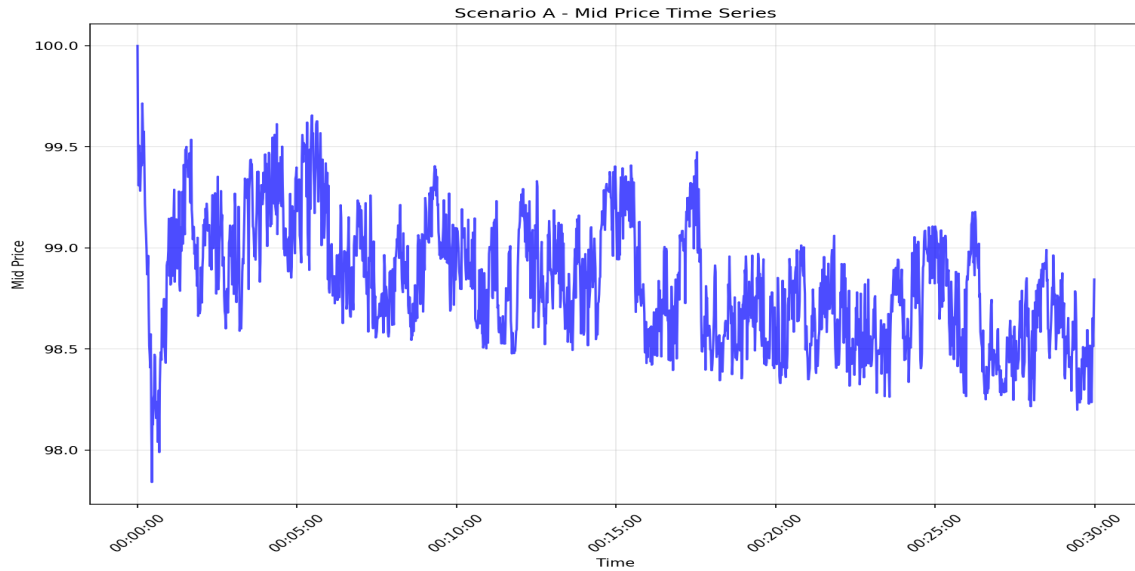
Scenario Composition

Scenario	Noise Traders	Market Makers	Momentum Traders	Total
A	100	0	0	100
B	80	20	0	100
C	80	0	20	100

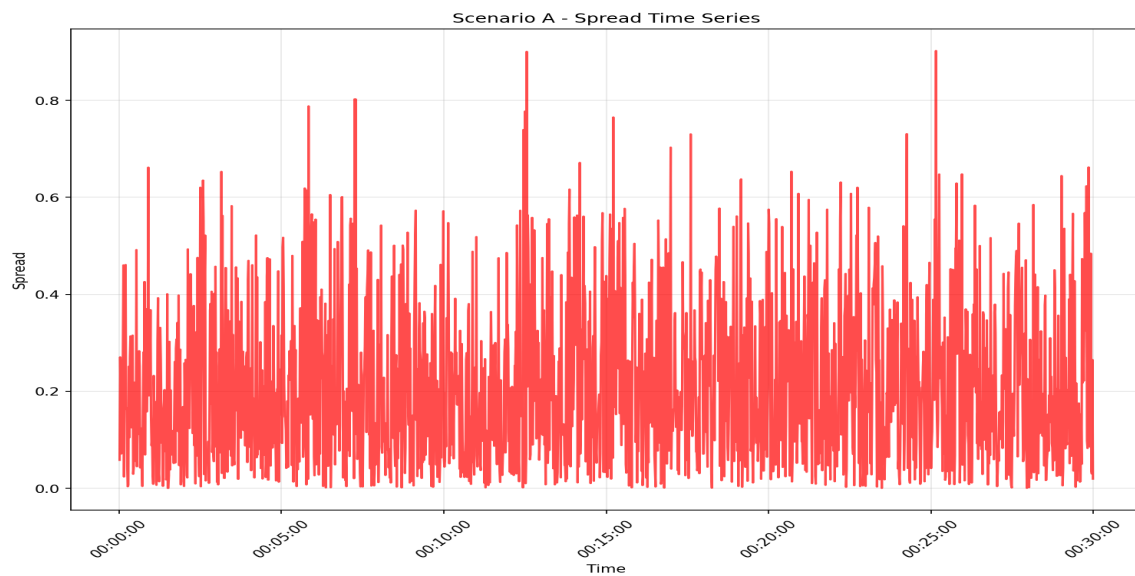
Scenario A Analysis

This scenario consists of 100 Noise Traders, 0 Market Makers, and 0 Momentum Traders. Noise traders act randomly without regard to market conditions. Market makers provide liquidity by quoting both bid and ask prices. Momentum traders follow trend signals based on moving averages.

Mid-Price Time Series



Spread Time Series



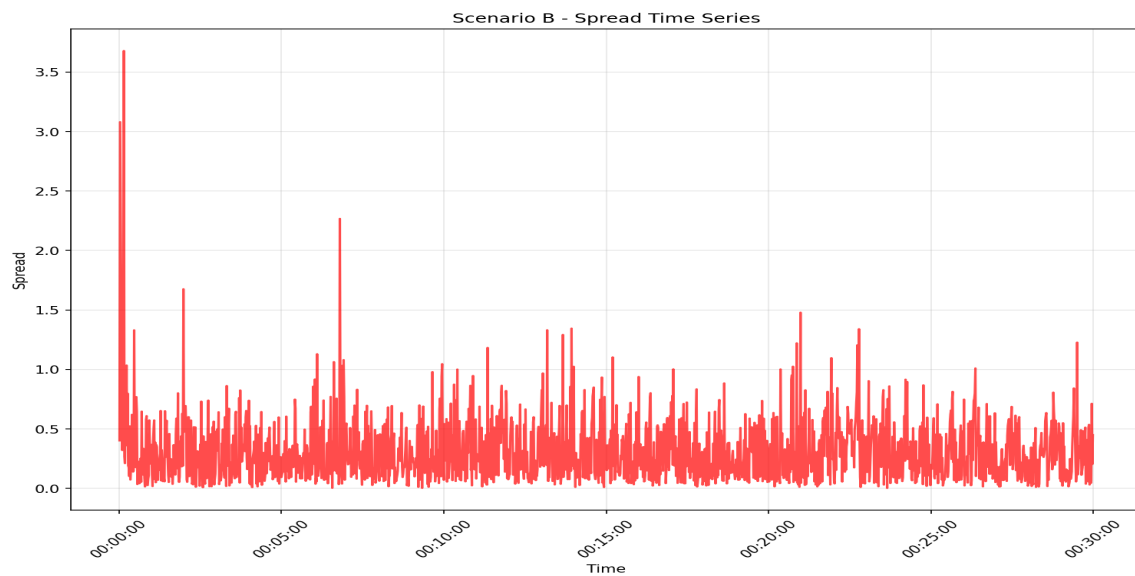
Scenario B Analysis

This scenario consists of 80 Noise Traders, 20 Market Makers, and 0 Momentum Traders. Noise traders act randomly without regard to market conditions. Market makers provide liquidity by quoting both bid and ask prices. Momentum traders follow trend signals based on moving averages.

Mid-Price Time Series



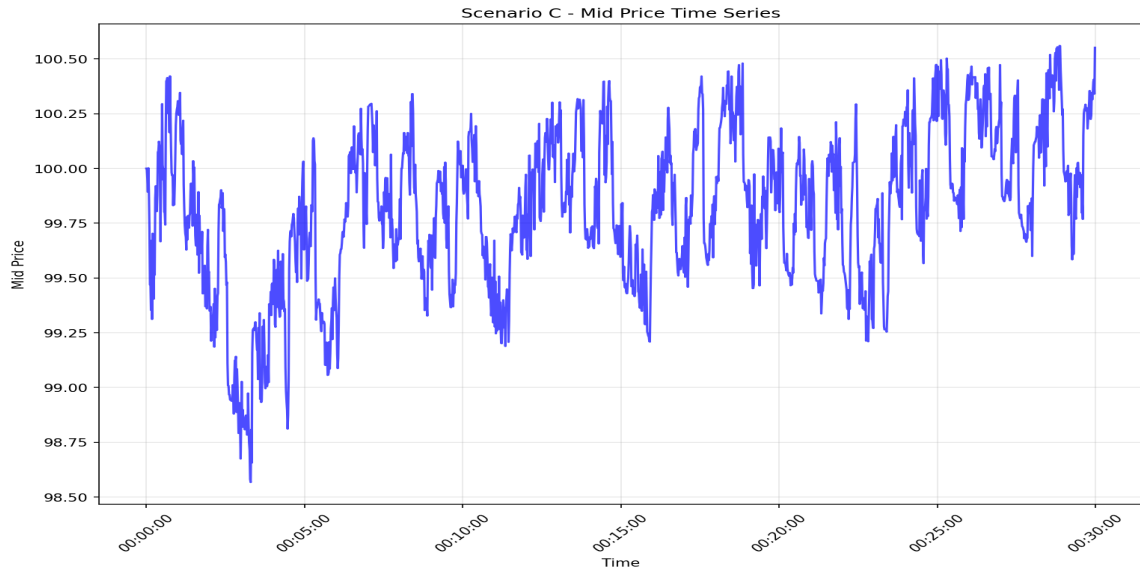
Spread Time Series



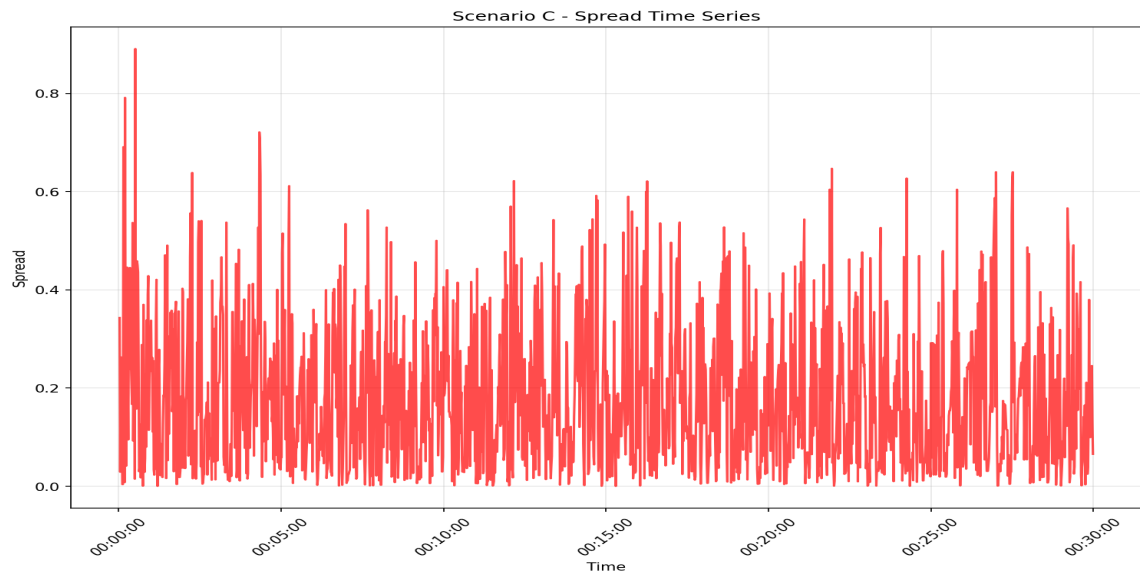
Scenario C Analysis

This scenario consists of 80 Noise Traders, 0 Market Makers, and 20 Momentum Traders. Noise traders act randomly without regard to market conditions. Market makers provide liquidity by quoting both bid and ask prices. Momentum traders follow trend signals based on moving averages.

Mid-Price Time Series



Spread Time Series

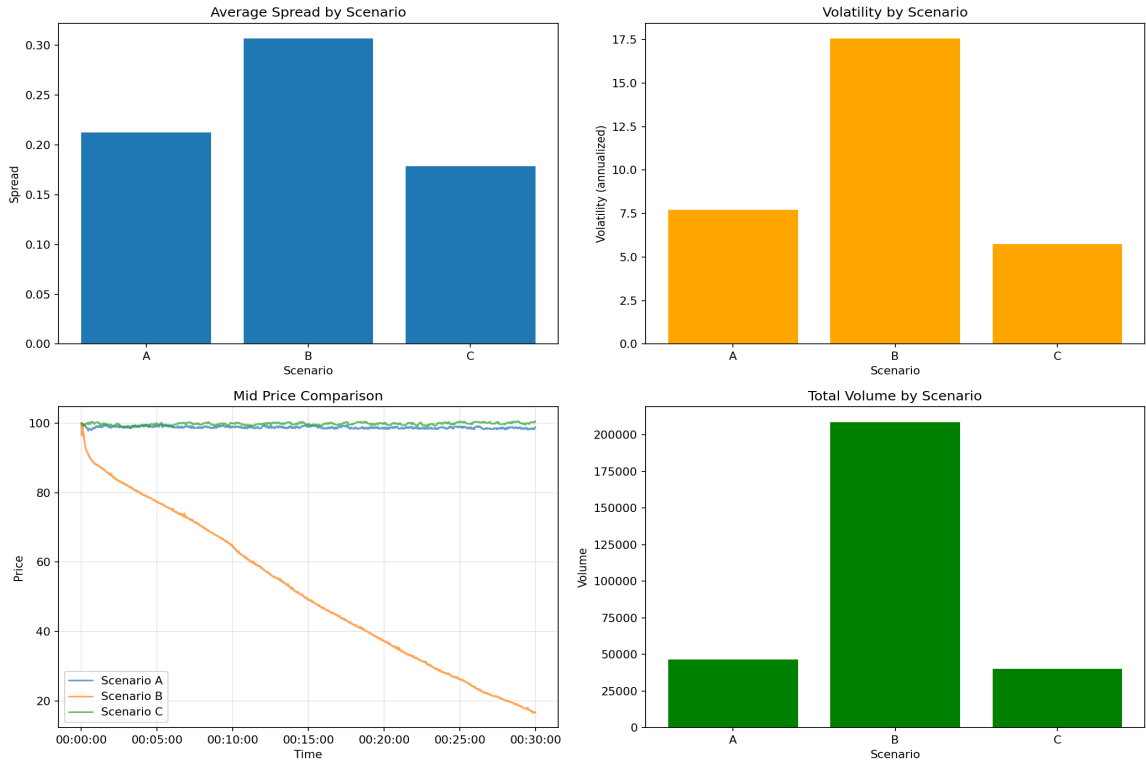


Scenario Comparison

Metric	Scenario A	Scenario B	Scenario C
Avg Spread	0.2126	0.3070	0.1785
Volatility	7.7058	17.5692	5.7397
VWAP	98.82	50.43	99.83
Total Volume	46378	208754	40170

Avg Trade Size	3.01	2.83	3.02
Price Range	2.56	90.20	2.06
Num Trades	15432	15432	15432

Visual Comparison



Interpretation & Analysis

Why Scenario B stabilizes prices: Market makers provide continuous liquidity by posting both bid and ask quotes. This constant presence of orders tightens the bid-ask spread and dampens price volatility. Their inventory management helps absorb temporary order imbalances. **Why Scenario C destabilizes prices:** Momentum traders create positive feedback loops. When prices rise, they buy more, pushing prices higher. When prices fall, they sell, accelerating declines. This herding behavior amplifies trends and increases volatility. **Why crashes occur naturally:** Even without explicit crash mechanisms, markets can experience crashes due to: 1. Liquidity evaporation when many traders act simultaneously 2. Margin calls or portfolio rebalancing creating forced selling 3. Psychological thresholds triggering stop-loss orders 4. Information cascades where traders imitate others' actions **Key Insights:** 1. Liquidity providers (market makers) are essential for market stability 2. Trend-following strategies can destabilize markets 3. Random trading creates baseline volatility 4. Market microstructure significantly

impacts price formation 5. Agent heterogeneity is crucial for realistic market simulation **Policy**

Implications: 1. Market makers should be incentivized during volatile periods 2. Circuit breakers can prevent momentum-driven crashes 3. Transaction taxes may reduce noise trading 4. Transparency in order books improves market efficiency